IE437 · DATA-DRIVEN DECISION MAKING AND CONTROL — LECTURE 0

Data-Driven Decision Making and Control

Deciding well under uncertainty — one question, three axes

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The map — how every chapter fits one story

01

The one question

Every lecture this term is the same question, asked again with one more thing made harder.

What this course is about — in one sentence

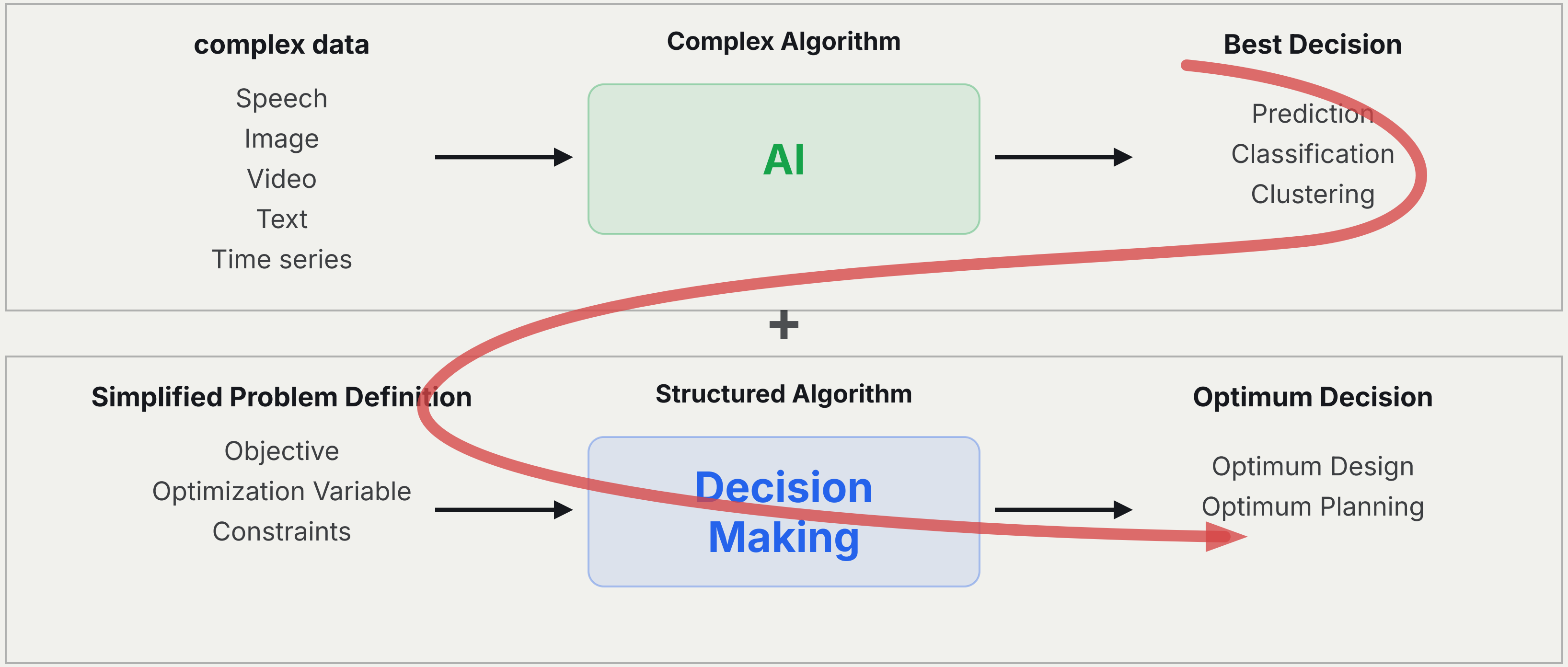
How do we make **good decisions under uncertainty?**

That is the entire course. Every lecture is this question, asked again with one more thing taken away — the model made uncertain, the dynamics unknown, the world shared with rivals.

Two disciplines answer it differently, and this course lives where they meet — **decision-centric AI**: use data and learning to make high-dimensional decisions that classical optimisation alone could not reach.

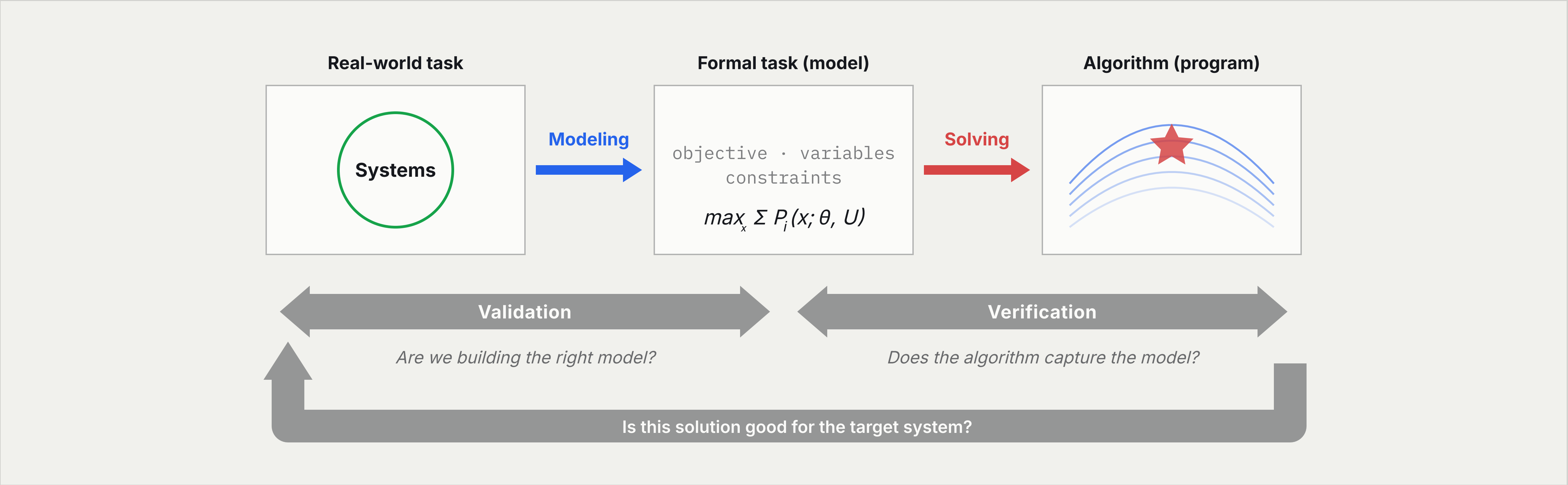
AI, decision making, and what joins them

TWO PIPELINES, AND WHAT JOINS THEM



Decision-centric AI. Feed what AI learns from data into the decision problem, and high-dimensional decisions become reachable that neither could reach alone.

From a real system to a solved problem



Formulation and solution are two different arrows, policed by two different questions. **Data helps on the left arrow too** — a model fitted to data represents the world more faithfully, not merely faster.

Two kinds of uncertainty — name them once

Everything we model is uncertain in one of two ways. The distinction recurs all term.

- **Aleatoric** (statistical) uncertainty — the world is genuinely stochastic. Repeating the experiment gives different outcomes. No amount of data removes it; we can only *characterise* it, as a distribution.
- **Epistemic** (systematic) uncertainty — we simply *don't know* the model: its parameters, its state, its dynamics. Data **reduces** it.

The whole course is a campaign against epistemic uncertainty.

Learning the parameter (Ch 2), the structure (Ch 3), the function (Ch 4), the design map (Ch 5–6), the dynamics (Ch 8–11) — while *respecting* the aleatoric uncertainty that no data can erase.

02

The three axes

Any decision problem sits somewhere along three independent axes. The course is a tour of this cube.

The map — three axes of every decision problem

AXIS	ONE END	THE OTHER END
Stages	static — a single decision	dynamic — a sequence
Model	model-based — handed to us	data-driven — learned
Decision makers	single agent — optimisation	many agents — a game

We traverse the cube in a deliberate order:

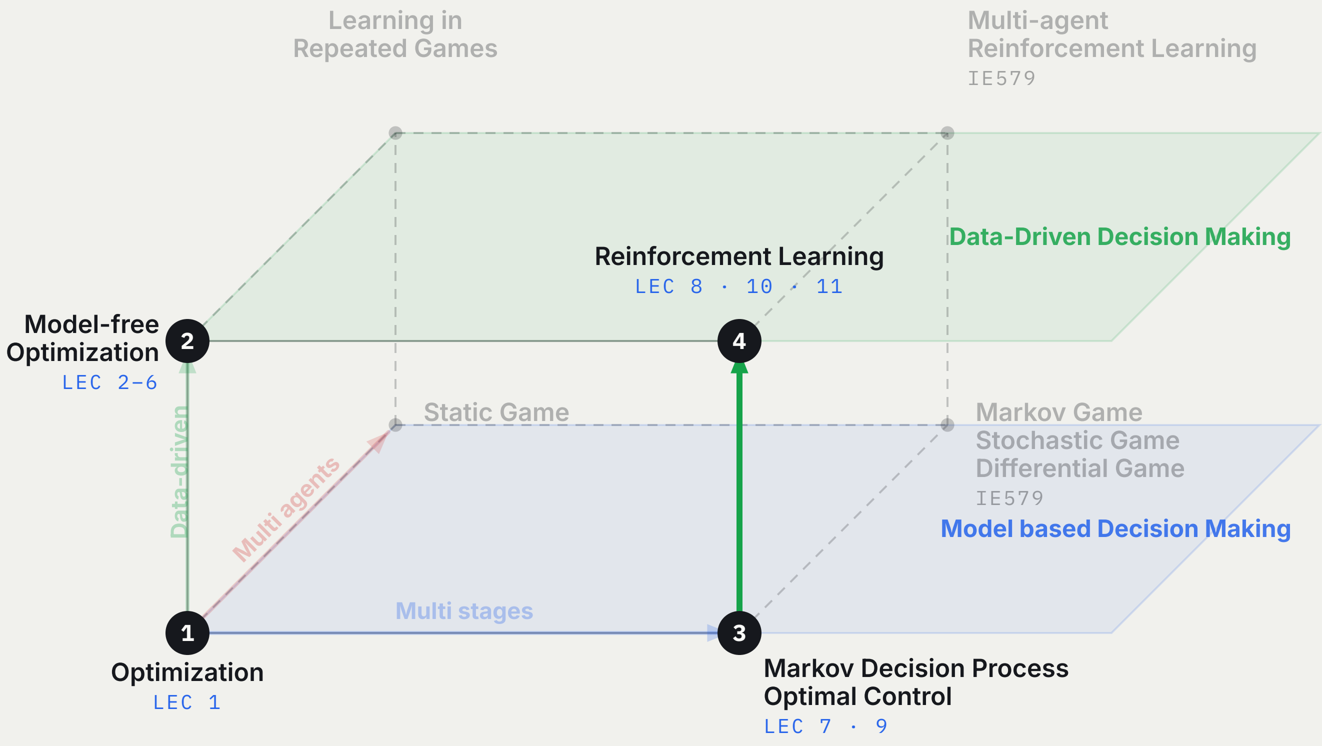
static → dynamic, **model-based → data-driven**, and then **interactive → offline**.

Each move is one bundle of lectures. None of the machinery is arbitrary — it is whatever that move requires.

The cube, and the route this course takes through it

THE CUBE — EVERY LECTURE, ONE MAP

STEP 4 / 6



④ **Reinforcement learning.** The data-driven axis again — but now **two** unknowns, reward *r* and transition *P*. The count doubles. *Lectures 8, 10, 11.*

The origin is plain optimisation. Step through the numbered route — **step ④ is the one to watch**, where the data-driven axis is crossed a second time and the count of unknowns doubles.

Axis 1 — static becomes dynamic

	SINGLE AGENT	MULTI AGENT
Static	optimisation	static game
Dynamic	dynamic optimisation / control	dynamic game

Static — one decision. You choose x once; the world does not move. This is classical optimisation (Lecture 1) and its uncertain, data-driven cousins (Lectures 2–6).

Dynamic — a sequence. Each decision reshapes the state the next one faces. This is the realm of MDPs, optimal control and reinforcement learning (Lectures 7–11).

The single optimisation splits into time — and the value function is born.

Axis 2 — model-based becomes data-driven

the deepest axis, and the course's namesake

Two ways to know the world you decide in:

- **Model-based** — the objective, constraints, transition and dynamics are *given*. You compute the optimum. *(Lectures 1, 7, 9)*
- **Data-driven** — you have only *data*. You must learn enough of the world to act: a belief (Ch 2–4), a surrogate (Ch 5–6), a value or a policy (Ch 8, 10, 11).

THE MOVE THAT DEFINES THE COURSE

Take a model-based method, delete the model, and replace it with data.

$$\text{optimisation} \to \text{Bayesian optimisation} \qquad \text{DP} \to \text{value-based RL} \qquad \text{optimal control} \to \text{policy-based RL}$$

Almost every lecture *pair* in this course is one instance of this single move.

Axis 3 — one decision maker becomes many

	OPTIMISATION	GAME
single objective	one optimum x^*	—
coupled objectives	—	equilibrium (Nash, ...)

When other agents — also optimising, also learning — share the world, the very notion of "optimal" changes. There is no single best action independent of what the others do. The solution concept shifts from an **optimum** to an **equilibrium**.

This course stops before that axis. Crossing it is the subject of the follow-on course, **IE579 Game Theory and Multi-Agent Reinforcement Learning** — and the cube above shows exactly which face it takes over. Naming the axis here is what makes the boundary of this course legible.

03

The doubling

The model axis is crossed twice — and the second crossing is not the first one again.

Crossing the model axis, twice

FIRST CROSSING · LECTURE 1 → 2

The static world. Delete the model and exactly **one** object goes missing:

$$\min_x \; f(x) \quad \text{s.t.} \quad g(x) \le 0$$

The answer is a **belief over f** — a posterior, a surrogate, a generative inverse.

SECOND CROSSING · LECTURE 7 → 8

The dynamic world. Delete the model and **two** objects go missing at once:

$$Q^*(s,a) = \sum_{s'} P(s' \mid s,a)\big[R(s,a,s') + \gamma \max_{a'} Q^*(s',a')\big]$$

Reward r **and** transition P — the count doubles.

This is why RL is not merely optimisation done with data.

Three answers to the same doubling

Value-based
Ch 8 · learn Q from samples — r and P are *fused* and neither is modelled

→

Policy-based
Ch 10 · output the action — the dynamics *cancel* in the gradient

→

Model-based
Ch 11 · estimate P *explicitly*, then plan with it

Part IV of the course is exactly these three answers, in that order. Read the whole of it as three different ways to survive the same doubling.

Each answer pays a different price: fusing loses the ability to plan ahead; cancelling costs sample efficiency; estimating invites model bias. No answer dominates — which is why all three are taught.

04

The two lineages

Inside the dynamic, single-agent cell there are two distinct intellectual origins — and this course teaches both, then their data-driven children.

Inside the dynamic cell — two scientific parents

	LINEAGE A · OR / DYNAMIC PROGRAMMING	LINEAGE B · CONTROL THEORY
MODEL-BASED (ORIGIN)	**MDP & Dynamic Programming** LECTURE 7	**Optimal Control / Planning** LECTURE 9
DATA-DRIVEN (EXTENSION)	**Value-Based RL** LECTURE 8	**Policy-Based RL** LECTURE 10

Value-based and policy-based RL are not two methods — they are two heritages.

- **Value-based RL** is *dynamic programming with the model deleted* — discrete, value-centric, Bellman's operations research.
- **Policy-based RL** is *optimal control with the dynamics deleted* — continuous, control-law-centric, Pontryagin and Kalman's control theory.

Lecture 11 then *reunites* them: learn the model back, and let value and policy methods teach each other.

The same claim, in two grids

pp. 57–58 of the source deck — where Lectures 7 to 10 come from

Zoom into the dynamic, single-agent cell and split it twice: by **action space** and by **time**. Handed a model, the four boxes are the classical theories.

MODEL-BASED	FINITE ACTION SPACE	INFINITE ACTION SPACE
discrete time	discrete-time MDP · $P(s_{t+1} \mid s_t, a_t)$	discrete-time dynamic system · $x_{t+1} = f(x_t, u_t)$
continuous time	continuous-time MDP · $P(s_{t+h} \mid s_t, a_t)$	continuous-time dynamic system · $\dot{x}_t = f(x_t, u_t)$

Now delete the model and the same grid names the data-driven methods:

MODEL-FREE	FINITE ACTION SPACE	INFINITE ACTION SPACE
discrete time	value-based reinforcement learning	policy-based reinforcement learning

The two-lineage claim is not an interpretation imposed on the course — it is this pair of grids. **The left column becomes Lectures 7 and 8; the right column becomes Lectures 9 and 10.** Both chapters redraw this figure at their own opening, and the empty continuous-time row is where the HJB equation of Lecture 9 lives.

05

The single spine

Read the whole course as a slow stripping-away of what you were handed.

Each chapter removes one given

WHAT YOU WERE HANDED — AND WHEN IT IS TAKEN AWAY

Lecture 8

objective f what "good" means	not in play
the right to query f may I evaluate a new x?	not in play
reward r the per-step objective	unknown — only samples
dynamics P / f how the state moves	unknown — only samples
sole decision maker nobody else optimises	handed to us

VALUE-BASED RL

OBJECTS STILL TO BE LEARNED **2**

Walk the lectures and watch the ledger empty. The counter on the right is the argument of the previous act: it reads **1** for the whole static half, and **2** from Lecture 8 onward.

What the course is for — four systems

Every method in this course was built against a real system. These four run through the lectures, and the numbers are the professor's own.

SYSTEM	WHAT IS DECIDED	METHOD	RESULT
Wind farm	where to place N turbines, in a shared wake	graph network for power, then layout optimisation *(Lec 1, 5, 11)*	1.5% MAPE prediction; total power 7.5 → 20.0 over the optimisation
Traffic signals	the green split at every intersection	offline meta Bayesian optimisation *(Lec 4)*	waiting vehicles 408.5 → **395.5** in Hangzhou, 920.1 → **859.7** in Manhattan
Semiconductor furnace	the heater work-set, every step	learned dynamics plus model-predictive control *(Lec 11)*	2.03 °C average error at a 100-step horizon
Combinatorial design	a chip placement, a route, a schedule	neural combinatorial optimisation	the search space is $> 10^{90,000}$ states — beyond Go's 10^{360}

None of these is a benchmark. Each is a system where a wrong decision costs power, time or yield — which is why the course spends its first six lectures on *how to state the problem* before it spends any on how to learn one.

Extension toward practical RL

p. 59 and p. 66 of the source — the professor's own roadmap

Textbook reinforcement learning assumes a simulator you may query without limit. Three extensions close the gap to a real plant, and this course takes all three.

1 · Model-based RL
learn the dynamics, then plan *(Lecture 11)*

→

2 · Meta RL
one policy across tasks whose dynamics and objectives change

→

3 · Offline RL
learn from a log, with no interaction at all *(Lecture 12)*

WHY OFFLINE RL — IN THE SOURCE DECK'S OWN WORDS

"Industrial systems typically do not have simulators, and it can be prohibitive to learn a policy by directly interacting with the real system. **Deriving a policy using previously collected data (offline data) is preferable.**"

Meta RL is named here and left to the follow-on material; model-based and offline RL become Lectures 11 and 12. That is why this course ends where it does — not because the cube runs out, but because **this is the axis a real deployment actually runs into.**

The course, as a route through the map

- **Part I — the given world.** Ch 1: classical optimisation, the atom of every later method.
- **Part II — the uncertain world.** Ch 2 Bayesian statistics (belief as a distribution) → Ch 3 Bayesian networks (structured belief, plus decisions) → Ch 4 Bayesian optimisation (act on an unknown function).
- **Part III — design from data alone.** Ch 5–6 data-driven design optimisation: surrogate (forward) and generative (inverse).
- **Part IV — the world that unfolds.** Ch 7 MDP & DP → Ch 8 value-based RL ‖ Ch 9 optimal control → Ch 10 policy-based RL → Ch 11 model-based RL.
- **Part V — when you cannot even try.** Ch 12 offline RL: the same dynamic, data-driven cell, with the right to interact withdrawn.

Appendix: a probability review — the toolbox underneath all of it.

06

Closing

One story, told in chapters — and a map to keep in view for the rest of the term.

One question — decide well under uncertainty — asked along three axes, answered by two lineages.

And unfolded by removing, one at a time, everything you were given.

Let us begin — Lecture 1.

Keep the three-axis map in mind: at the start of every lecture we mark exactly where we stand on it, and which assumption we are about to give up.